POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Electronic Devices and Circuits

Semester: Fall

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs..

Candidates are required to give their answers in their own words as far as practicable.

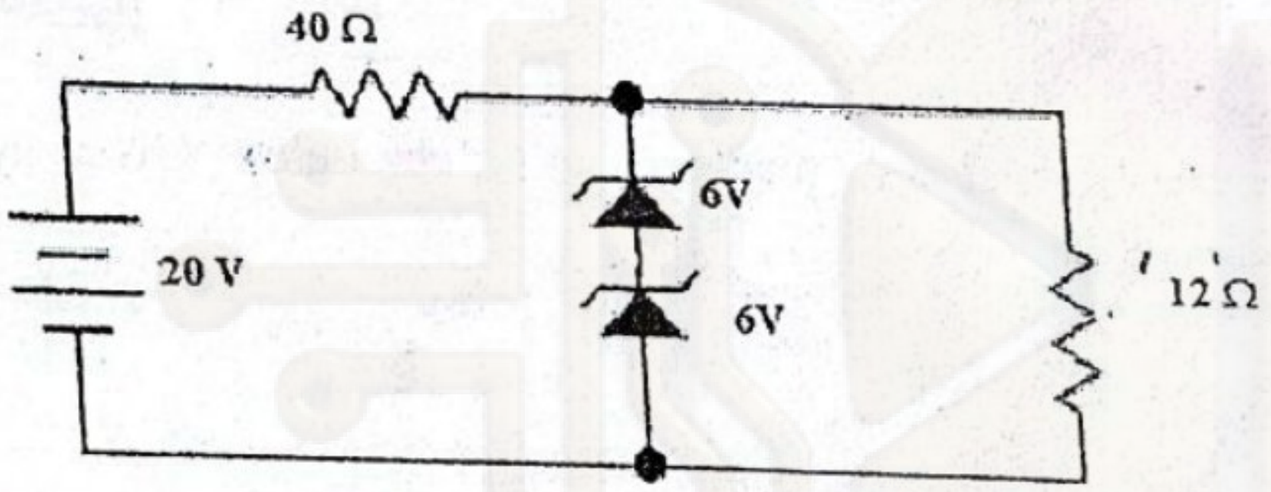
The figures in the margin indicate full marks.

Attempt all the questions.

a) Draw V-I characteristics curve of pn junction diode. Distinguish between a zener breakdown and avalanche breakdown. 8

b) Differentiate between intrinsic and extrinsic semiconductor. A silicon diode has saturation current of 5 nanoampere at room temperature. What is the saturation current at 100 degree centigrade. 7

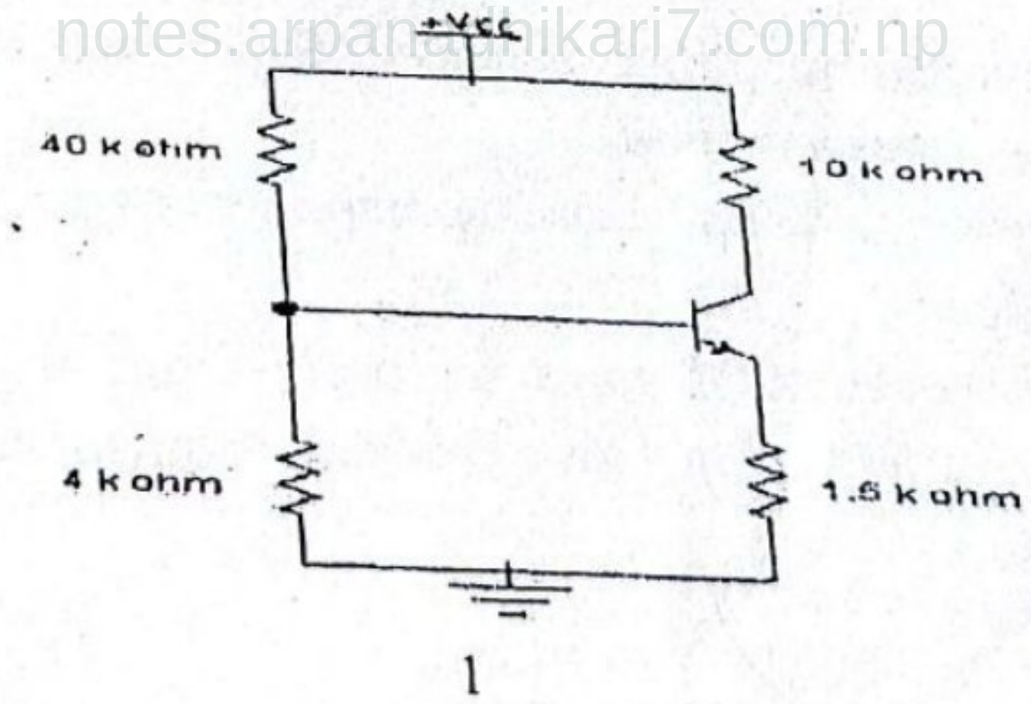
a) Calculate the current through load resistor and zener diode. 7



b) For the transistor amplifier shown below given that Vcc=22 V. 8

i. Draw dc load line.

ii. Operating point.



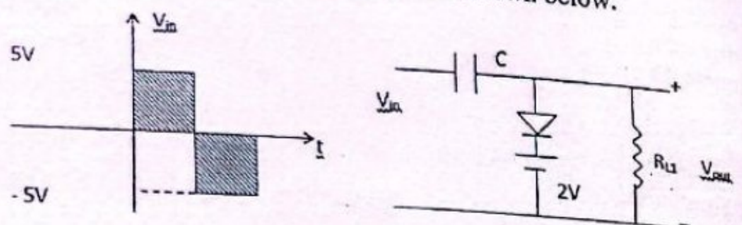
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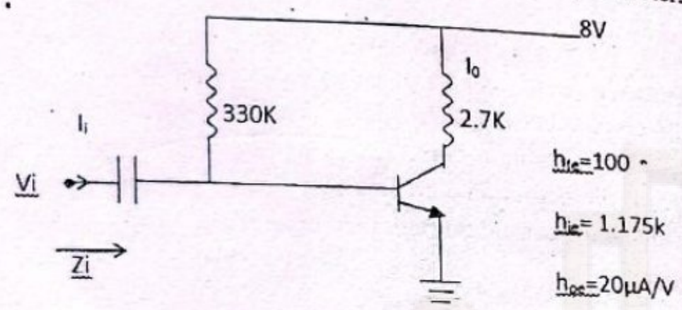
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3. a) Explain the input characteristics of Common emitter transistor. Also derive the relationship between α, β, and γ. 7

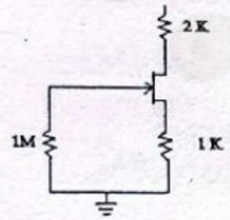
b) Sketch the output waveform for the circuit shown below. 8



4. a) Find A_{vs}, A_v, Z_0, Z_{in} of following Ckt. Diagram using h- parameter. 8



b) Determine I_{DQ} and V_{DSQ} for the circuit given below. Given I_{DSS} = 8 mA, V_P = -8 V. 7



5. a) Darlington-pair amplifier is called Super beta why? Derive the necessary expression for it. 7

b) Draw transformer coupled class A amplifier and derive the expression of its efficiency. 8

6. a) Derive the equation of gains for positive and negative feedback. Explain the effect of negative feedback configuration in amplifier characteristics. 7

b) For the op-amp based wein bridge oscillator if the component values are: 8

R=5.1kΩ, C=1nF R_3=12kΩ and R_4=5.1kΩ

i. Determine whether the circuit will oscillate or not.

ii. Obtain the output frequency.

7. Write short notes on: **(Any two)** 2×5

a) Crystal Oscillator.

b) Emitter Follower.

c) Input impedance.

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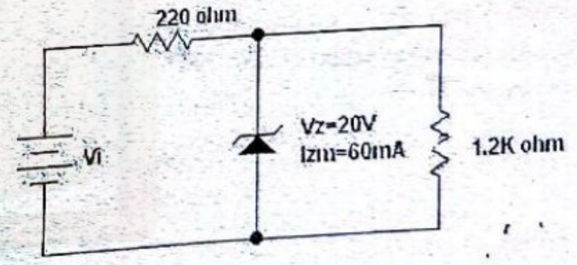
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POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2015
Programme: BE Full Marks: 100
Course: Electronic Device and Circuits Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What happens when a forward biased diode is suddenly reverse biased? Explain with necessary diagrams. 5
   b) What do you mean by dynamic resistance of diode? Show that dynamic resistance of diode $r_D=\eta V_T/I$. 5
   c) Determine the range of values of Vin that will maintain the zener diode in the "ON" state. Also find the maximum power that can be dissipated to the diode. 5

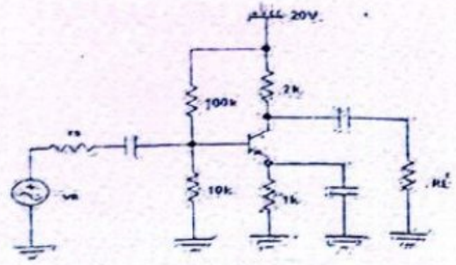


2. a) Explain the working principle of NPN transistor with necessary diagram. Show that $I_C=\beta I_B+(\beta+1)I_{CBO}$. 8
   b) For the circuits shown below, draw the dc load line and find the operating point. Also find the stability S of the circuit. (Assume silicon transistor with $\beta=100$) 7

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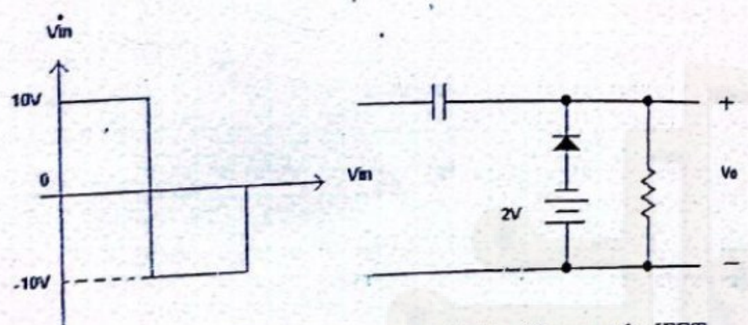
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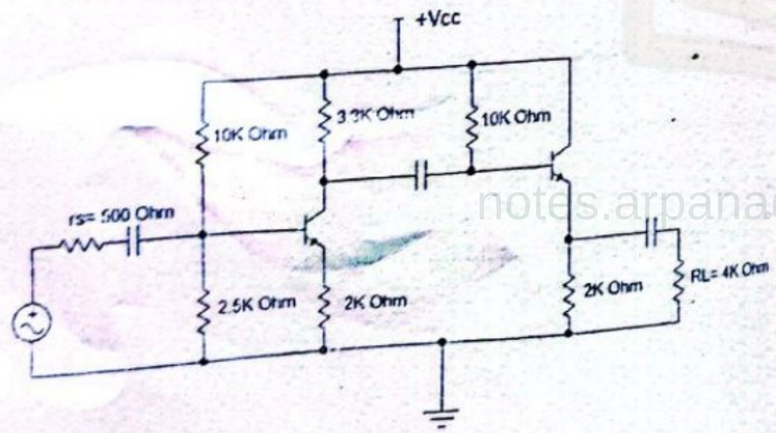
3. a) Explain the operation of Centre-Tap full wave rectifier. Find its efficiency. ripple factor and PIV. 8

b) Draw the output waveform (assume silicon diode). 7



4. a) Compare BJT and FET. Explain how pinch-off occurs in JFET. 7

b) Find the overall voltage gain of cascaded transistor amplifier as shown in figure below. Given $r_{e1}=r_{e2}=25\Omega$ and $\beta1=\beta2=110$. 8



5. a) What do you mean by class A amplifier. Find the maximum efficiency of class A amplifier with resistive load. 2+6

b) What do you mean by feedback amplifier? Show that negative feedback improves the stability of an amplifier. 2+5

6. a) What are the conditions for oscillations? Draw a Wien-bridge Oscillator and explain its operation. 5

b) What do you mean by Operational Amplifier? Compare the characteristics of Ideal and practical Operational Amplifiers. 5

c) Design an Operational Amplifier circuit to produce the following Output. 5

$V_O= 3V_1+2V_2-0.5V_3$. Where V_1, V_2, V_3 are inputs and V_O is an output.

7. Write short notes on: **(Any two)** 2×5

a) Effect of Bypass Capacitor on voltage gain of CE amplifier

b) Darlingtone Pair

c) Cross over Distortion

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Electronic Device and Circuits

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

सुगम स्टेशनरी साप्लायर्स एण्ड फोटोकपी सर्भिस
बालकुमारी, ललितपुर ९८४१५१९५९२
NCIT College

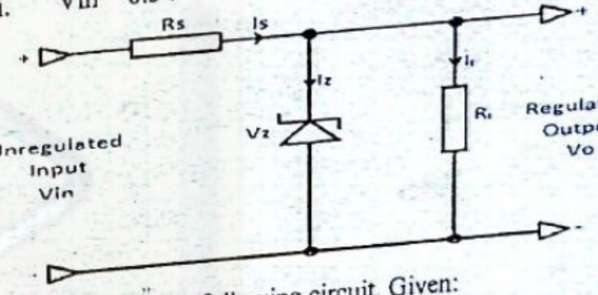
Attempt all the questions.

a) What is semiconductor material? Explain the importance of semiconductor material on electronic Device with examples. 8

b) If Vz = 8V, Rs = 100Ω, R_L = 990Ω. Find Vo, I_L, Iz, if, 7

i. Vin = 12v

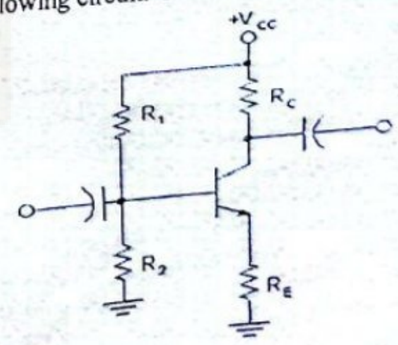
ii. Vin = 8.5V



7

2. a) Find Q point for the following circuit. Given:

Vcc = 16V
R_1 = 62K
R_2 = 9.1K
R_C = 3.9K
R_E = 1K
β = 80



4

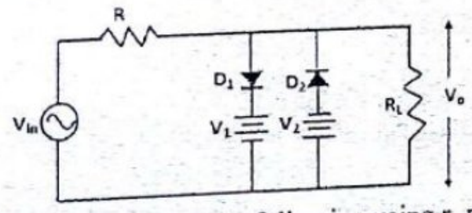
b) Explain BJT as an amplifier. 4

c) Compare CB, CE and CC configurations. 8

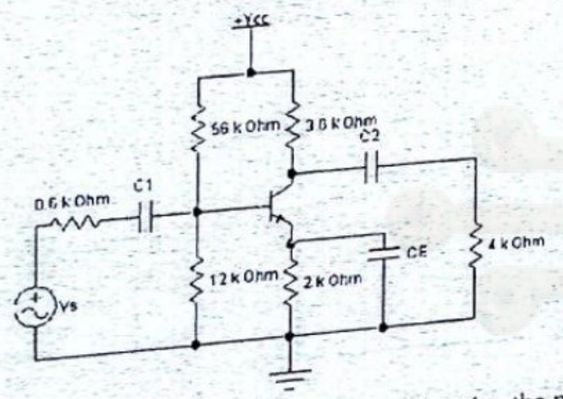
1. a) Define rectification. Explain that the efficiency of full wave rectifier is 1

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b) double of the half wave rectifier.
Draw the wave form of voltage across R_L. Assume Vin = 10 sin wt Volts. Also explain the operation to support your waveform. 7



4. a) For the given CE amplifier, find the following using r_e model 8
i. input impedance
ii. output impedance
iii. voltage gain
(Assume $\beta = 90$, $V_{CC} = +19V$)



b) Darlington-pair amplifier is called Super beta why? Derive the necessary expression for it. 7

5. a) Derive efficiency of ~~Transformer coupled~~ Class A amplifier? Explain cross over distortion in class B push pull amplifier. 8

b) Derive the equation of gains for positive and negative feedback. Explain the effect of negative feedback configuration in amplifier characteristics. 7

6. a) State Ideal characteristics of op-amp? Derive closed loop gain of inverting, non inverting and integrator amplifier? 8

b) Calculate the component values of Wein Bridge Oscillator to be used in oscillator to vary frequency from 100 Hz to 10 kHz in two stages. 7

7. Write short notes on: (Any two) 2×5
a) Slew Rate and CMRR
b) Clamping and Clipping Circuit
b) FET Parameters

POKHARA UNIVE...

Level: Bachelor
Programme: BE
Course: Electronic Devices and Circuits

Semester: Spring
Year
Full Marks: 100
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Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
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Attempt all the questions.

a) Differentiate between Diffusion Capacitance and Transition Capacitance. "Transition capacitance is inversely proportional to the square root of the reverse voltage applied across it." Verify this statement. 3+5

b) Explain the Zener diode as a voltage regulator with its I-V characteristics. 7

a) Explain the BJT as a switch. 5

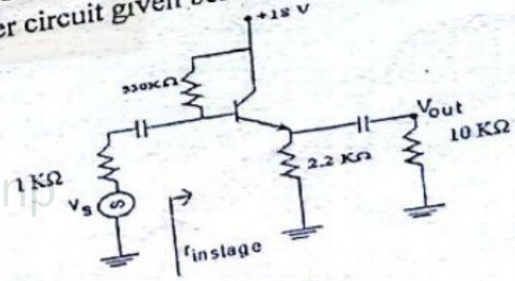
b) What are the Delay time and Rise time? Briefly explain the Reach Through or Punch Through effect. 2+3

c) An NPN silicon transistor having a nominal β of 100 is to be used in a CE configuration with Vcc = 15 V. The Q point is to be Ic = 4mA and VcE = 12V. Now design the circuit diagram. 5

3. a) What do you mean by voltage regulation? Explain with the block diagram. 7

b) What is pinch off voltage? Compare the JFET with BJT. 2+6

4. a) For the amplifier circuit given below find: 8



i. r_{in} ii. r_{out} iii. A_v iv. A_{vs}
Where the symbols have their usual meanings. (Given $\beta = 100$, $r_e = 20\Omega$).

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